INCH-POUND
MIL-PRF-19500/630B
22 December 1997
SUPERSEDING
MIL-PRF-19500/630A
2 May 1997

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, FIELD EFFECT RADIATION HARDENED TRANSISTOR, P-CHANNEL, SILICON, TYPES 2N7389, 2N7390, 2N7389U, AND 2N7390U JANTXV, R, AND F AND JANS, R, AND F

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for a P-channel, Radiation Hardened, enhancement mode, MOSFET, power transistor. Two levels of product assurance are provided for each device type as specified in MIL-PRF-19500, with avalanche energy ratings (E_{AS}) and maximum avalanche current (I_{AS}).

1.2 Physical dimensions. See figure 1 (T0-205AF) and figure 3 (LCC).

1.3 Maximum ratings. Unless otherwise specified, T_C = +25° C.

Type 4/	Min $V_{(BR)DSS}$ V_{GS} = 0 V I_D =-1.0 mA dc	P_T 1/	P_T 1/ T_A = +25° C (free air)	V_{GS}	I_{D1} 2/	I_{D2} 2/ T_C = +100° C	T_J and T_{STG}
	V dc	W	W	V dc	A dc	A dc	°C
2N7389	-100	25	0.8	±20	-6.5	-4.1	-55 to +150
2N7390	-200	25	0.8	±20	-4.0	-2.4	-55 to +150

Type 4/	I_S	I_{DM}	V_{ISO} 70,000 feet altitude	Max $r_{DS(on)}$ 1/ V_{GS} =-12 V dc I_D = I_{D2}		$R_{\Theta JC}$ max	E_{AS}	I_{AS}
				T_J =+25° C	T_J =+150° C			
	A dc	A(pk)		ohms	ohms	°C/W	mJ	A dc
2N7389	-6.5	-26		0.30	0.60	5	165	-6.5
2N7390	-4.0	-16		0.80	1.68	5	171	-4.0

1/ Derated linearly by 0.2 W/° C for T_C > +25° C; $P_T = \frac{T_J \text{ max} - T_C}{R_{\Theta JC}}$

2/ $I_D = \sqrt{\frac{T_J max - T_c}{(R_{\Theta JC}) \times (R_{DS}(on) at T_{Jmax})}}$

3/ I_{DM} = 4 x I_{D1}; I_{D1} as calculated by footnote 2/.

4/ Electrical characteristics, ratings, and conditions for "U" suffix devices are identical to the corresponding non "U" suffix devices, unless otherwise specified.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad St., Columbus, OH 43216-5000, by using the addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5961

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1.4 Primary electrical characteristics. Unless otherwise specified, T_C = +25° C.

Type 2/	Min $V_{(BR)DSS}$ V_{GS} = 0 V I_D =-1.0 mA dc	$V_{GS(th)1}$ $V_{DS} \geq V_{GS}$ I_D =-1.0 mA dc	I_{DSS} max V_{GS} = 0 V	Max $r_{DS(on)1}$ 1/ V_{GS} =-12 V; $I_D = I_{D2}$
			V_{DS} = 80% of rated V_{DS}	T_J = +25° C
	V dc	V dc	µA dc	ohms
		Min Max		
2N7389	-100	-2.0 -4.0	-25	0.30
2N7390	-200	-2.0 -4.0	-25	0.80

1/ Pulsed (see 4.5.1).
2/ Electrical characteristics, ratings, and conditions for "U" suffix devices are identical to the corresponding non "U" suffix devices, unless otherwise specified.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

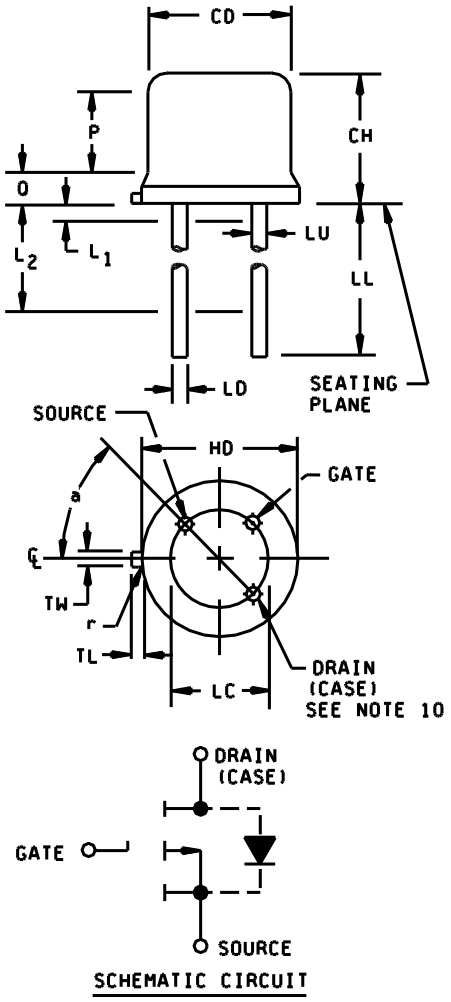
2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated specification. The individual item requirements shall be in accordance with MIL-PRF-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as defined in MIL-PRF-19500.

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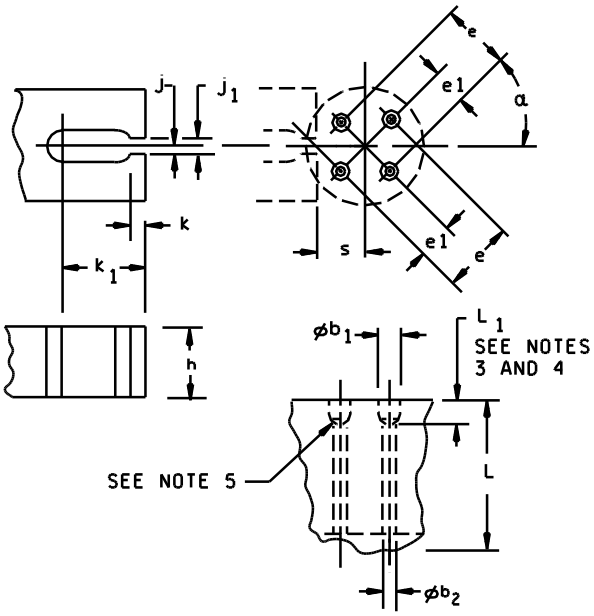
Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CD	.305	.335	7.75	8.51	
CH	.160	.180	4.07	4.47	
HD	.335	.370	8.51	9.40	
LC	.200 TP		5.08 TP		6
LD	.016	.021	0.41	0.53	7,8
LL	.500	.750	12.70	19.05	7,8
LU	.016	.019	0.41	0.48	7,8
L_1		.050		1.27	7,8
L_2	.250		6.35		7,8
P	.100		2.54		5
Q		.050		1.27	4
r		.010		0.25	9
TL	.029	.045	0.74	1.14	3
TW	.028	.034	0.71	0.86	2
α	45°		45°		6

NOTES:
1. Dimensions are in inches. Metric equivalents are given for general information only.
2. Beyond radius (r) maximum, TW shall be held for a minimum length of .011 (0.028 mm).
3. Dimension TL measured from maximum HD.
4. Outline in this zone is not controlled.
5. Dimension CD shall not vary more than .010 (0.25 mm) in zone P. This zone is controlled for automatic handling.
6. Leads at gauge plane .054 +.001, -.000 (1.37 +0.03, -0.00 mm) below seating plane shall be within .007 (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC. The device may be measured by direct methods or by the gauge and gauging procedure shown on figure 2.
7. LU applies between L_1 and L_2. LD applies between L_2 and LL minimum. Diameter is uncontrolled in L_1 and beyond LL minimum.
8. All three leads.
9. Radius (r) applies to both inside corners of tab.
10. Drain is electrically connected to the case.
11. In accordance with ANSI Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 1. Physical dimensions for TO-205AF.

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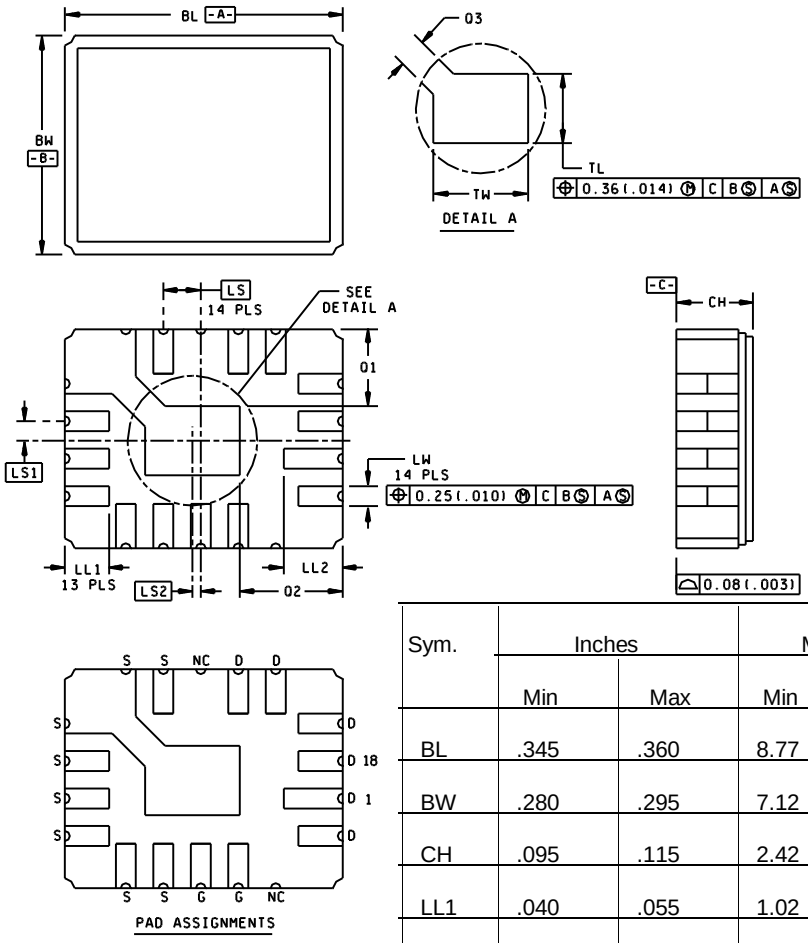
Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
ϕb_1	.0595	.0605	1.511	1.537	
ϕb_2	.0325	.0335	0.826	0.851	
e	.1995	.2005	5.067	5.093	
e_1	.0995	.1005	2.527	2.553	
h	.150 Nominal		3.81 Nominal		
j	.0175	.0180	0.444	0.457	
j_1	.0350	.0355	0.889	0.902	
k	.009	.011	0.23	0.28	
k_1	.125 Nominal		3.18 Nominal		
L	.372	.378	9.45	9.60	
L_1	.054	.055	1.37	1.40	
s	.182	.199	4.62	5.05	2
α	44.90°	45.10°	44.90°	45.10°	



NOTES:
1. Dimensions are in inches. Metric equivalents are given for general information only.
2. The location of the tab locator within the limits indicated will be determined by the tab and flange dimensions of the device being checked.
3. Gauging procedure. The device being measured shall be inserted until its seating plane is .125 ±.010 (3.18 ±0.25 mm) from the seating surface of the gauge. A force of 8 ±.5 ounces shall then be applied parallel and symmetrical to the device's cylindrical axis. When examined visually after the force application (the force need not be removed), the seating plane of the device shall be seated against the gauge. The use of a pin straightener prior to insertion in the gauge is permissible.
4. Gauging plane.
5. Drill angle.

FIGURE 2. Gauge for lead and tab location.

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Sym.	Inches		Millimeters	
	Min	Max	Min	Max
BL	.345	.360	8.77	9.14
BW	.280	.295	7.12	7.49
CH	.095	.115	2.42	2.92
LL1	.040	.055	1.02	1.39
LL2	.055	.065	1.40	1.65
LS	.050 BSC		1.27 BSC	
LS1	.025 BSC		0.635 BSC	
LS2	.008 BSC		0.203 BSC	
LW	.020	.030	0.51	0.76
Q1	.105 REF		2.67 REF	
Q2	.120 REF		3.05 REF	
Q3	.045	.055	1.15	1.39
TL	.070	.080	1.78	2.03
TW	.120	.130	3.05	3.30

NOTES:
1. Dimensions are in inches. Metric equivalents are given for information only.
2. In accordance with ANSI Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 3. Physical dimensions for LCC.

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3.3 Interface requirements and physical dimensions. The Interface requirements and physical dimensions shall be as specified in MIL-PRF-19500, and on figures 1 and 3 herein. The preferred measurements used herein are the metric units. However, this transistor was designed using inch-pound units of measurement. In case of conflicts between the metric and inch-pound units, the inch-pound units shall be rule.

3.3.1 Lead material and finish. Lead material shall be Kovar. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition requirement (see 6.2).

3.3.2 Internal construction. Multiple chip construction shall not be permitted.

3.4 Marking. Marking shall be in accordance with MIL-PRF-19500. At the option of the manufacturer, marking of country of origin may be omitted from the body of the transistor, but shall be retained on the initial container.

3.5 Electrostatic discharge protection. The devices covered by this specification require electrostatic protection.

3.5.1 Handling. MOS devices must be handled with certain precautions to avoid damage due to the accumulation of static charge. The following handling procedures shall be followed:

a. Devices shall be handled on benches with conductive handling devices.

b. Ground test equipment, tools, and personnel handling devices.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent, if practical.

g. Care shall be exercised, during test and troubleshooting, to apply not more than maximum rated voltage to any lead.

h. Gate must be terminated to source. $R \leq 100$ k, whenever bias voltage is to be applied drain to source.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in 4.4.2 and 4.4.3.

3.8 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.2).

4. VERIFICATION

4.1 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3)

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500.

4.2.1 Group E inspection. Group E inspection shall be in accordance with table III herein.

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4.3 Screening (JANS and JANTXV levels). Screening shall be in accordance with table IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement	
	JANS levels	JANTXV levels
1/	Thermal response 4.5.3	Thermal response 4.5.3
1/	Method 3470, see 4.5.5	Method 3470, see 4.5.5
1/	Gate stress test 4.5.4	Gate stress test 4.5.4
9 1/	I_{GSS1}, I_{DSS1}, subgroup 2 of table I herein	Subgroup 2 of table I herein
10	Method 1042, test condition B	Method 1042, test condition B
11	I_{GSS1}, I_{DSS1}, $r_{DS(on)1}$, $V_{GS(th)1}$ subgroup 2 of table I herein. ΔI_{GSS1} = ±20 nA dc or ±100% of initial value, whichever is greater. ΔI_{DSS1} = ±25 µA dc or ±100% of initial value, whichever is greater.	I_{GSS1}, I_{DSS1}, $r_{DS(on)1}$, $V_{GS(th)1}$ subgroup 2 of table I herein.
12	Method 1042, test condition A, t = 240 hours	Method 1042, test condition A
13	Subgroup 2 and 3 of table I herein. ΔI_{GSS1} = ±20 nA dc or ±100% of initial value, whichever is greater. ΔI_{DSS1} = ±25 µA dc or ±100% of initial value, whichever is greater. $\Delta r_{DS(on)(1)}$ = ±20% of initial value. $\Delta V_{GS(th)1}$ = ±20% of initial value.	Subgroup 2 of table I herein. ΔI_{GSS1} = ±20 nA dc or ±100% of initial value, whichever is greater. ΔI_{DSS1} = ±25 µA dc or ±100% of initial value, whichever is greater. $\Delta r_{DS(on)(1)}$ = ±20% of initial value. $\Delta V_{GS(th)1}$ = ±20% of initial value.

1/ Shall be performed anytime before screen 10.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein. Electrical measurements (end points) and delta requirements shall be in accordance with the inspections of table I, group A, subgroup 2 herein.

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4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIa (JANS) and table VIb (JANTX and JANTXV) of MIL-PRF-19500, and as follows. Electrical measurements (end points) shall be in accordance with the applicable inspections of table I, group A, subgroup 2 herein.

4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
3	1051	Condition G, 100 cycles.
4	1042	The heating cycle shall be 30 seconds minimum.
5	1042	Condition B; V_{GS} = 100 percent of rated; T_A = +175° C; t = 24 hours, or T_A = +150° C; t = 48 hours (manufacturers option).
5	1042	Condition A; V_{DS} = 100 percent of rated; T_A = +175° C, t = 120 hours, or T_A = +150° C, t = 240 hours (manufacturers option).
6	3161	See 4.5.2

4.4.2.2 Group B inspection, table VIb (JANTX and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Condition
2	1051	Test condition G, 25 cycles.
3	1042	The heating cycle shall be 30 seconds minimum.
3	2037	Test condition A.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500, and as follows. Electrical measurements (end points) shall be in accordance with the applicable inspections of table I, group A, subgroup 2 herein.

4.4.3.1 Group C inspection, table VII of MIL-PRF-19500.

Subgroup	Method	Condition
2	2036	Test condition E (Applicable to T0-205AF only).
6	1042	The heating cycle shall be 30 seconds minimum.

4.5 Methods of inspection. Methods of inspection shall be as specified in appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurements shall be as specified in section 4 of MIL-STD-750.

4.5.2 Thermal resistance. Thermal resistance measurements shall be performed in accordance with method 3161 of MIL-STD-750. The maximum limit of $R_{\Theta JC(max)}$ shall be 5.0° C/W. The following parameter measurements shall apply:

a. Measuring current (I_M): 10 mA.
b. Drain heating current (I_H): 1.0 A minimum for T0-205AF devices, 1.3 A minimum for LCC devices.
c. Heating time (t_H): Steady-state (see MIL-STD-750, method 3161 for definition).
d. Drain-source heating voltage (V_H): 25 V minimum for T0-205AF devices, 15 V minimum for LCC devices.
e. Measurement time delay (t_{MD}): 30 to 60 μs maximum.
f. Sample window time (t_{SW}): 10 μs maximum.

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4.5.3 Thermal impedance ($Z_{\Theta JC}$ measurements). The $Z_{\Theta JC}$ measurements shall be performed in accordance with MIL-STD-750, method 3161. The maximum limit (not to exceed figure 4, thermal impedance curves and the group A, subgroup 2 limits) for $Z_{\Theta JC}$ in screening (table IV of MIL-PRF-19500) shall be derived by each vendor by means of statistical process control. When the process has exhibited control and capability, the capability data shall be used to establish the fixed screening limit. In addition to screening, once a fixed limit has been established, monitor all future sealing lots using a random five piece sample from each lot to be plotted on the applicable X, R chart. If a lot exhibits an out of control condition, the entire lot shall be removed from the line and held for Engineering evaluation and disposition. This procedure may be used in lieu of an in line process monitor.

a. Measuring current (I_M): 10 mA.
b. Drain heating current (I_H): 1.0 A minimum for T0-205AF devices, 1.3 A minimum for LCC devices.
c. Heating time (t_H): 10 ms.
d. Drain-source heating voltage (V_H): 25 V minimum for T0-205AF devices, 15 V minimum for LCC devices.
e. Measurement time delay (t_{MD}): 30 to 60 μs maximum.
f. Sample window time (t_{SW}): 10 μs maximum.

4.5.4 Gate stress test.
a. V_{GS} = -24 V min.
b. t = 250 µs min.

4.5.5 Single pulse avalanche energy (E_{AS}).
a. Peak current (I_{AS}): I_{D1}.
b. Peak gate voltage (V_{GS}): -12 V.
c. Gate to source resistor (R_{GS}): $25 \leq R_{GS} \leq 200\ \Omega$.
d. Initial case temperature: +25° C +10° C, -5° C.
e. Inductance: $(2\ E_{AS}/(I_{AS})^2)((V_{BR} - V_{DD})/V_{BR})$ mH minimum.
f. Number of pulses to be applied: 1 pulse minimum.
g. Supply voltage V_{DD} = -50 V, or -25 V for 100 V devices.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Thermal impedance 2/	3161	See 4.5.3	$Z_{\Theta JC}$		1.6	°C/W
Breakdown voltage, drain to source	3407	V_{GS} = 0 V dc, I_D = -1.0 mA; Bias condition C	$V_{(BR)DSS}$			
2N7389, 2N7389U				-100		V dc
2N7390, 2N7390U				-200		V dc
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$; I_D = -1.0 mA	$V_{GS(th)1}$	-2.0	-4.0	V dc
Gate current	3411	V_{GS} = ±20 V dc; V_{DS} = 0 V dc; Bias condition C	I_{GSS1}		±100	nA dc
Drain current	3413	V_{GS} = 0 V dc; V_{DS} = 80% of rated V_{DS}; Bias condition C	I_{DSS1}		-25	µA dc
Static drain to source on-state resistance	3421	V_{GS} =-12 V dc; Condition A; Pulsed (see 4.5.1); I_D = rated I_{D2} (see 1.3)	$r_{DS(on)1}$			
2N7389, 2N7389U					0.30	Ω
2N7390, 2N7390U					0.80	Ω
Static drain to source on-state resistance	3421	V_{GS} =-12 V dc; Condition A; Pulsed (see 4.5.1); I_D = rated I_{D1} (see 1.3)	$r_{DS(on)2}$			
2N7389, 2N7389U					0.35	Ω
2N7390, 2N7390U					0.92	Ω
Forward voltage (source drain diode)	4011	V_{GS} = 0 V dc; I_D = rated I_{D1}; Pulsed (see 4.5.1)	V_{SD}			
2N7389, 2N7389U					-3.0	V
2N7390, 2N7390U					-5.0	V

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3						
High temperature operation		T_A = +125° C				
Gate current	3411	Bias condition C; V_{GS} = ±20 V dc; V_{DS} = 0 V dc	I_{GSS2}		±200	nA dc
Drain current	3413	Bias condition C; V_{GS} = 0 V dc; V_{DS} = 80% of rated V_{DS}	I_{DSS3}		-0.25	mA dc
Static drain to source on-state	3421	V_{GS} = -12 V dc; Pulsed (see 4.5.1); I_D = rated I_{D2}	$r_{DS(on)3}$			
2N7389, 2N7389U					0.54	Ω
2N7390, 2N7390U					1.60	Ω
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$; I_D = -1.0 mA dc	$V_{GS(th)2}$	-1.0		V dc
Low temperature operation		T_A = -55° C				
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$; I_D = -1.0 mA dc	$V_{GS(th)3}$		-5.0	V dc
Subgroup 4						
Switching time test	3472	I_D = rated I_{D1}; V_{GS} = -12 V dc; Gate drive impedance = 7.5Ω; V_{DD} = 50% of $V_{(BR)DSS}$				
Turn-on delay time			$t_{d(on)}$			
2N7389, 2N7389U					30	ns
2N7390, 2N7390U					30	ns
Rise time			t_r			
2N7389, 2N7389U					50	ns
2N7390, 2N7390U					30	ns
Turn-off delay time			$t_{d(off)}$			
2N7389, 2N7389U					70	ns
2N7390, 2N7390U					75	ns
Fall time			t_f			
2N7389, 2N7389U					70	ns
2N7390, 2N7390U					65	ns

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4- continued						
Forward transconductance	3475	$I_D = I_{D2}$; V_{DD} = -15 V dc Pulsed (see 4.5.1)	g_{fs}			
2N7389, 2N7389U				2.5		S
2N7390, 2N7390U				2.5		S
Subgroup 5						
Safe operating area test (high voltage)	3474	See figures 5 and 6; t_p = 10 ms; V_{DS} = 80% of rated $V_{(BR)DSS}$				
Electrical measurements		See table I, group A, subgroup 2				
Subgroup 6						
Not applicable						
Subgroup 7						
Gate charge	3471	Condition B				
On-state gate charge			$Q_{g(on)}$			
2N7389, 2N7389U					45	nC
2N7390, 2N7390U					45	nC
Gate to source charge			Q_{gs}			
2N7389, 2N7389U					10.0	nC
2N7390, 2N7390U					10.0	nC
Gate to drain charge			Q_{gd}			
2N7389, 2N7389U					25	nC
2N7390, 2N7390U					25	nC
Reverse recovery time	3473	di/dt ≤ -100 A/µs; V_{DD} ≤ -50 V; $I_D = I_{D1}$	t_{rr}			
2N7389, 2N7389U					250	ns
2N7390, 2N7390U					400	ns

1/ For sampling plan, see MIL-PRF-19500.
2/ This test is required for the following end-point measurements only:

JANS - group B, subgroups 3 and 4; JANTXV - group B, subgroups 2 and 3; group C, subgroup 6; group E, subgroup 1

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TABLE II. Group D inspection.

Inspection 1/ 3/	MIL-STD-750		Symbol	Pre-irradiation limits		Post-irradiation limits		Post-irradiation limits		Unit
	Method	Conditions		R and F		R		F		
				Min	Max	Min	Max	Min	Max	
Subgroup 2		T_C = +25°C								
Steady-state total dose irradiation (V_{GS} BIAS) 2/	1019	V_{GS} = -12 V; V_{DS} = 0 V								
Steady-state total dose irradiation (V_{DS} BIAS) 2/	1019	V_{GS} = 0 V; V_{DS} = 80 percent of rated V_{DS} (preirradiation)								
End-point electricals										
Breakdown voltage, drain to source	3407	V_{GS} = 0 V; I_D = -1 mA; Bias condition C	$V_{(BR)DSS}$							
2N7389, 2N7389U				-100		-100		-100		V dc
2N7390, 2N7390U				-200		-200		-200		V dc
Gate to source 3/ voltage (threshold)	3403	$V_{DS} \geq V_{GS}$; I_D = -1 mA	$V_{GS(th)}$							
2N7389, 2N7389U				-2.0	-4.0	-2.0	-4.0	-2.0	-5.0	V dc
2N7390, 2N7390U				-2.0	-4.0	-2.0	-4.0	-2.0	-5.0	V dc
Gate current	3411	V_{GS} = -20 V; V_{DS} = 0 V; Bias condition C	I_{GSSF1}		-100		-100		-100	nA dc
Gate current	3411	V_{GS} = +20 V; V_{DS} = 0 V; Bias condition C	I_{GSSR1}		100		100		100	nA dc

See footnotes at end of table.

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TABLE II. Group D inspection - Continued.

Inspection 1/ 3/ 5/	MIL-STD-750		Symbol	Pre-irradiation limits		Post-irradiation limits 4/		Post-irradiation limits 4/		Units
	Method	Conditions		R and F		R		F		
				Min	Max	Min	Max	Min	Min	
Subgroup 2 - Continued.										
Drain current	3413	V_{GS} = 0 V; Bias condition C; VDS = 80% of rated V_{DS} (preirradiation)	I_{DSS}							
2N7389, 2N7389U					-25		-25		-25	µA dc
2N7390, 2N7390U					-25		-25		-25	µA dc
Static drain to source on-state voltage	3405	V_{GS} = -12 V; Condition A; Pulsed see 4.5.1; $I_D = I_{D2}$	$V_{DS(ON)1}$							
2N7389, 2N7389U					1.23		1.23		1.23	V dc
2N7390, 2N7390U					1.92		1.92		1.92	V dc
Forward voltage source drain diode	4011	V_{GS} = 0 V; $I_D = I_{D1}$; Bias condition C	V_{SD}							
2N7389, 2N7389U					-3.0		-3.0		-3.0	V
2N7390, 2N7389U					-5.0		-5.0		-5.0	V

1/ For sampling plan see MIL-PRF-19500.
2/ Separate samples shall be pulled for each bias.
3/ Group D qualification may be performed anytime prior to lot formation. Wafers qualified to these group D QCI requirements may be used for any other detail specification utilizing the same die design.
4/ The F designation represent devices which pass endpoints at both 100K and 300K rad (Si).
5/ At the manufacturer's option, group D samples need not be subjected to the screening tests, and may be assembled in it's qualified package or in any qualified package that the manufacturer has data to correlate the performance to the designated package.

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TABLE III. Group E inspection (all quality levels) for qualification only.

Inspection	MIL-STD-750		Qualification inspection
	Method	Conditions	
Subgroup 1			12 devices, c = 0
Temperature cycling	1051	Test condition G; 500 cycles	
Hermetic seal	1071		
Fine leak Gross leak			
Electrical measurements		See table I, group A, subgroup 2	
Subgroup 2 1/			12 devices, c = 0
Steady-state reverse bias	1042	Condition B; 1000 hours	
Electrical measurements		See table I, group A, subgroup 2	
Steady-state gate bias	1042	Condition A; 1,000 hours	
Electrical measurements		See table I, group A, subgroup 2	
Subgroup 3			
Not applicable			
Subgroup 4			12 devices, c = 0
Thermal resistance	3161	See 4.5.2	
Subgroup 5			
Not applicable			

1/ A separate sample for each test may be pulled.

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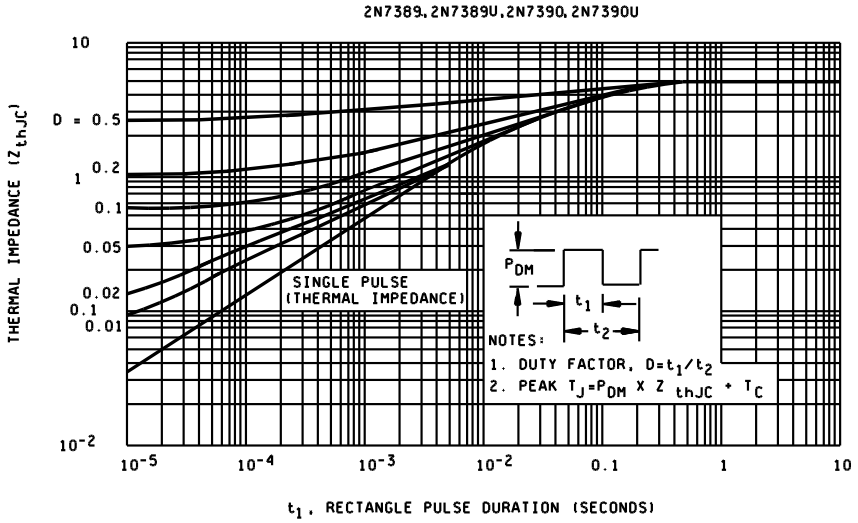


FIGURE 4. Thermal response curves.

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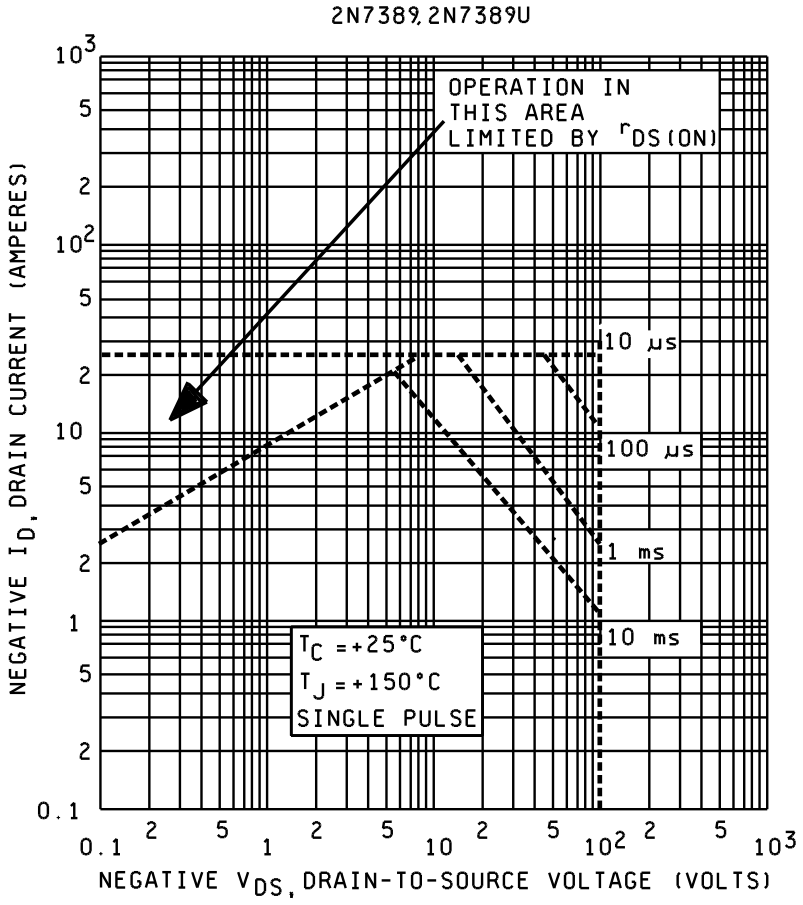


FIGURE 5. Safe operating area graphs.

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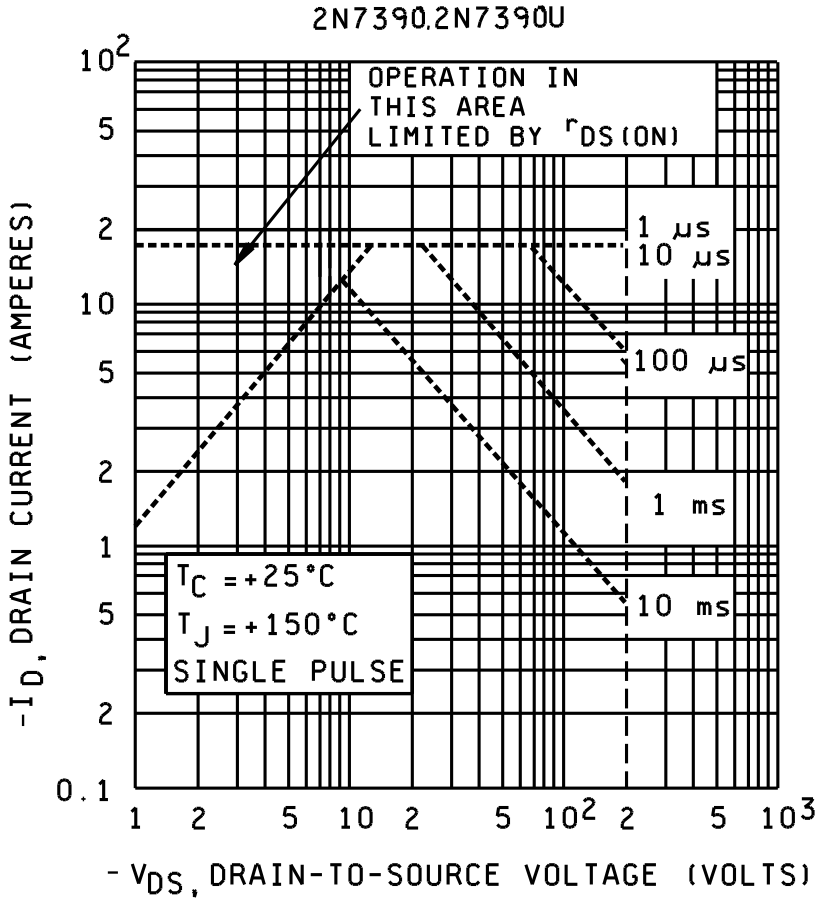


FIGURE 6. Safe operating area graphs.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead formation and finish as specified (see 3.3.1).

c. Product assurance level and type designator.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL-19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, DSCC-VQE, Columbus, OH 43216.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC
Air Force - 13, 19, 85, 99

Preparing activity:
DLA - CC

(Project 5961-1955)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-19500/630B	**2. DOCUMENT DATE** 22 December 1997

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, FIELD EFFECT RADIATION HARDENED TRANSISTOR, P-CHANNEL, SILICON, TYPES 2N7389, 2N7390, 2N7389U, AND 2N7390U JANTXV R AND F AND JANS R AND F

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Alan Barone	b. TELEPHONE (Include Area Code) (1) Commercial 614-692-0510 (2) AUTOVON 850-0510
c. ADDRESS (Include Zip Code) Defense Supply Center Columbus ATTN: DSCC-VAT Columbus, OH 43216-5000	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

DD Form 1426, OCT 89 Previous editions are obsolete 198/290

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